UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1.5.1

Tree-based Regression Models

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Machine Learning I

fic facultade de informática da coruña

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Regression

- Method that models the relationship between a dependent variable Y and independent variables Xi
  - Linear regression: try to approximate a function f(x) (supposedly linear) with a linear function f'(x)

$$f'(x) = w_0 + w_1 a_1(x) + w_2 a_2(x) + \ldots + w_n a_n(x)$$

  - It doesn't have to be linear:
    - Quadratic
    - Cubic
    - Quartic
    - Logarithmic
    - Logistics
    - etc.

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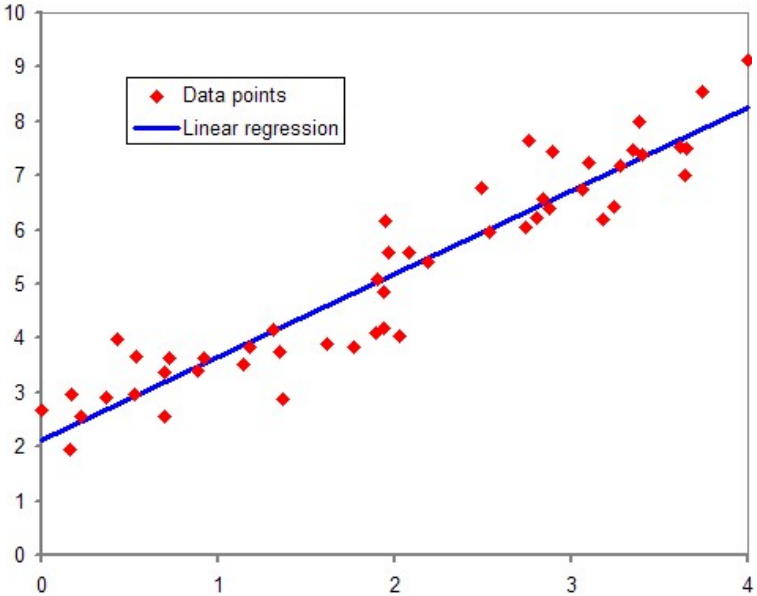
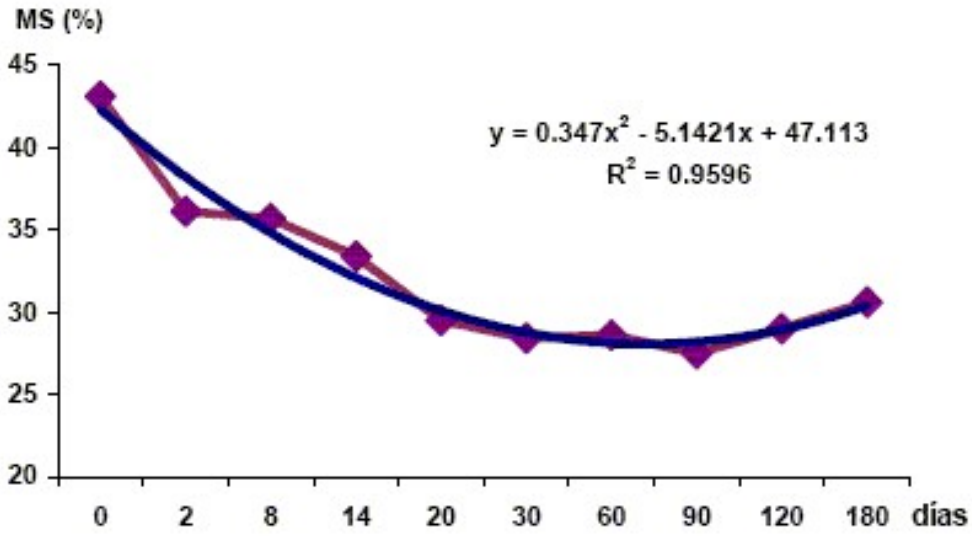
Regression

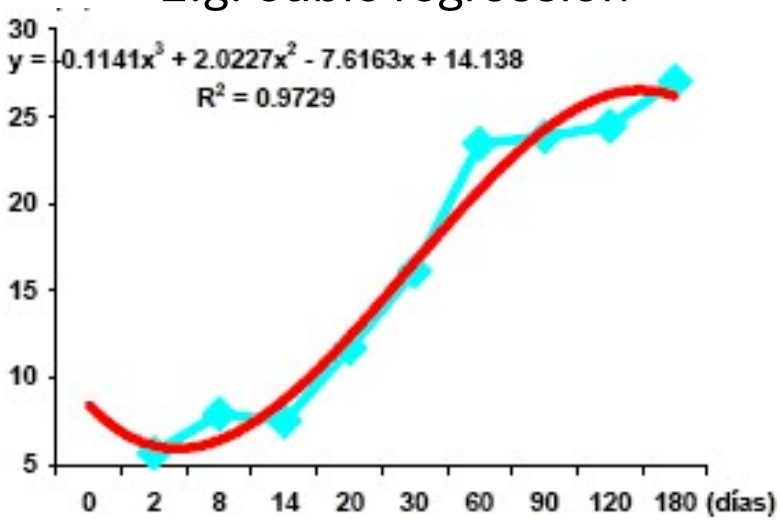
E.g. linear regression



E.g. quadratic regression



E.g. cubic regression



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Regression

- Objetive:
  - Minimize the error between the actual and the approximation values
- One possible error measure is the mean squared error over the total training set, D:

$$E = \frac{1}{N} \sum_{x \in D} (f(x) - f'(x))^2$$

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Regression

- In linear regression, the problem of defining the function

$$f'(x) = w_0 + w_1 a_1(x) + w_2 a_2(x) + \ldots + w_n a_n(x)$$

moves to a problem of defining the vector of weights w

- The vector w that minimizes the error function must be found:
  - Search problem in the weight space

- Known Statistical Methods
  - Least Squares

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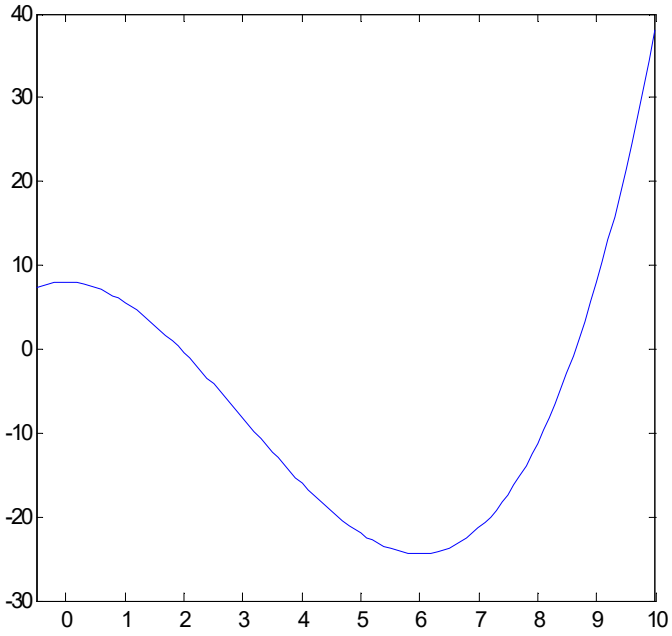
Regression

- Regression trees and regression models:
  - Linear regression is a global model
    - There is an equation applicable in the whole space (global model)
  - If data interacts in a non-linear and complicated way, getting a global linear model can be difficult
    - Example: Linear model?



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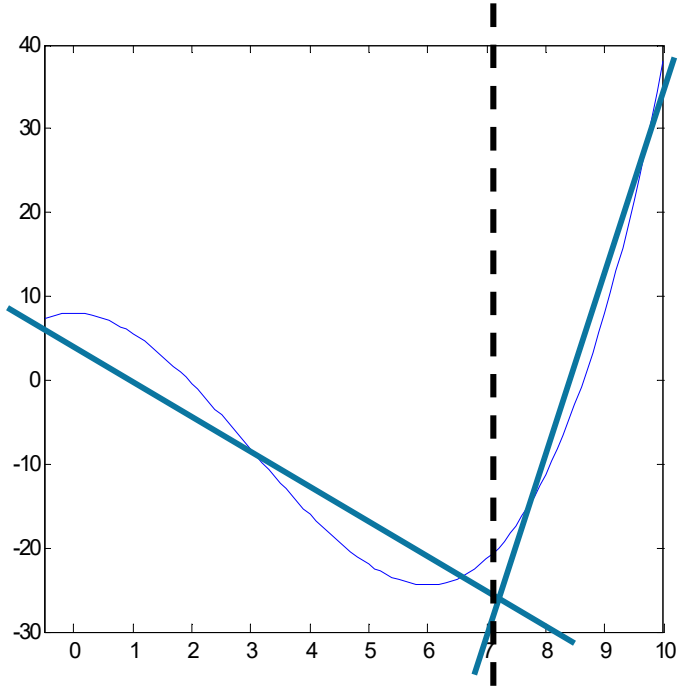
Regression

- Regression trees and regression models:
  - Alternative: split the space into smaller regions
    - In these regions, the interactions should be easier to model with a linear approach
    - In each region, create a simple model
      - Local model
    - Example:



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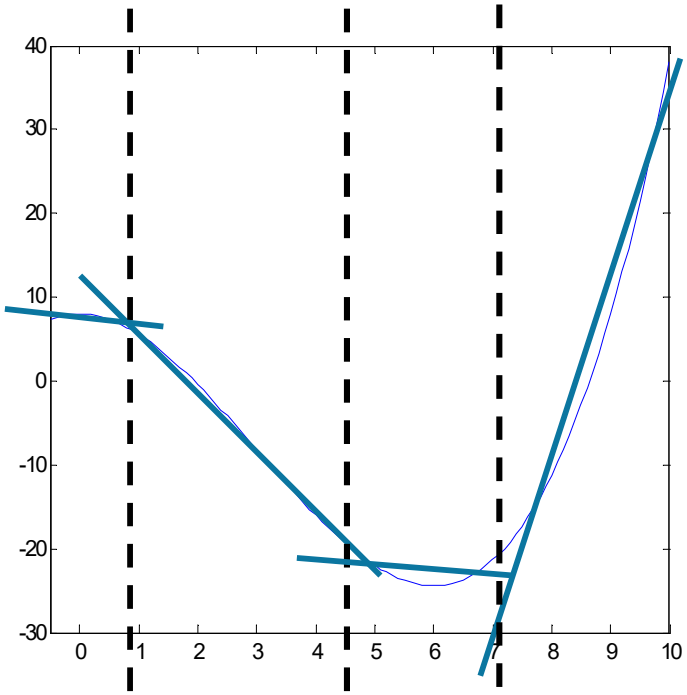
Regression

- Regression trees and regression models:
  - Alternative: partition space into smaller regions
    - Split regions recursively
    - Example:



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Regression

- Regression trees and regression models:
  - How to split the space?
    - Trees: similar to classification trees
  - What model to create in the regions?
    - Regression trees
    - Regression model trees

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Regression Trees

- Regression trees:
  - Generalization of decision trees to give continuous regression functions
  - Leaves:
    - Average value of the targets of the instances that arrive to that leaf
  - CART System (Classification And Regression Trees) (Breiman et al., 1984)
    - For both classification and regression trees
  - Works for all types of attributes
  - They are a particular case of regression model trees

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Regression Trees

- Regression trees:
  - They approximate the objective function by a piecewise **constant function**:
    - That is, the model in each region is a constant
    - If a set of c instances (x_i, y_i) arrive at a leaf node, the model in that leaf will be the average value of the values of the dependent variable of the instances that arrive at that node:

$$\frac{1}{c}\sum_{i=1}^{c} y_i$$

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Regression Trees

- Regression trees:
  - CART:
    - Impurity Criterion:
      - Instead of using the Gini index, the impurity criterion is the **sum of squares**
      - Therefore, the divisions that cause the greatest reduction in the sum of squares are chosen
    - Pruning:
      - When pruning the tree the measure used is the mean square error in the predictions made by the tree.

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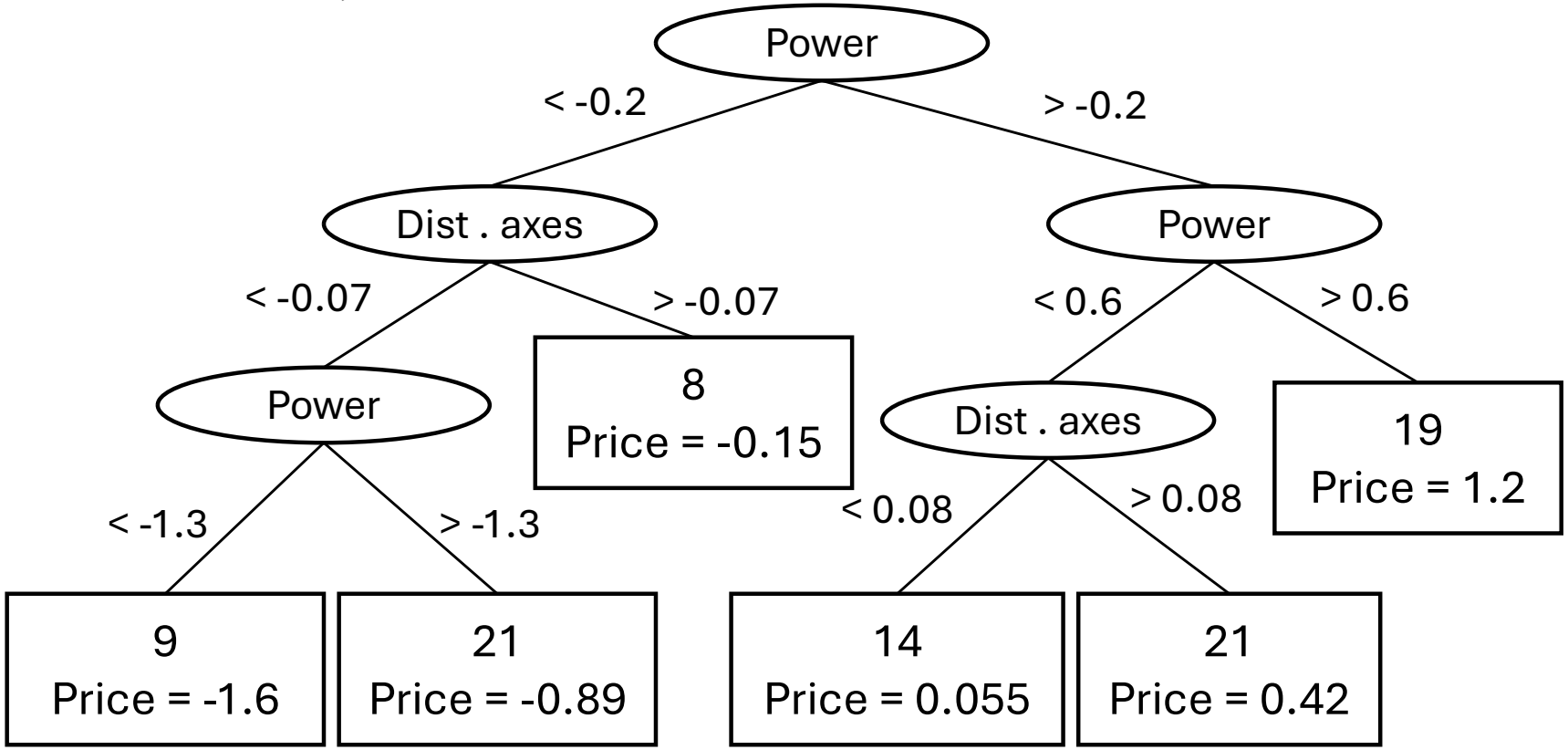
Regression Trees

- Regression trees:
  - Example: 1993 car prices:
    - All attributes, and the output variable, have been normalized
      - Mean = 0, and standard deviation = 1



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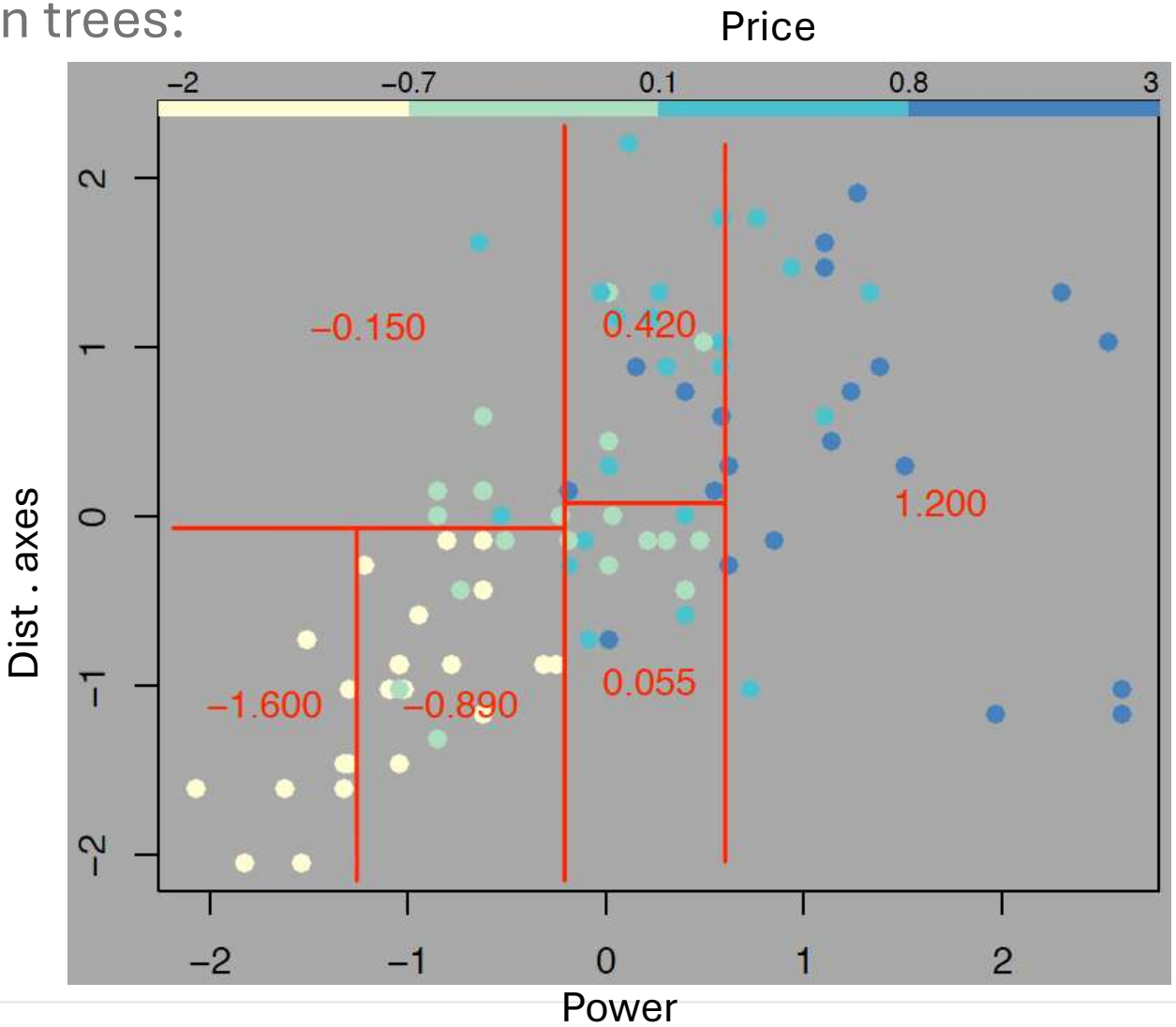
Regression Trees

- Regression trees:



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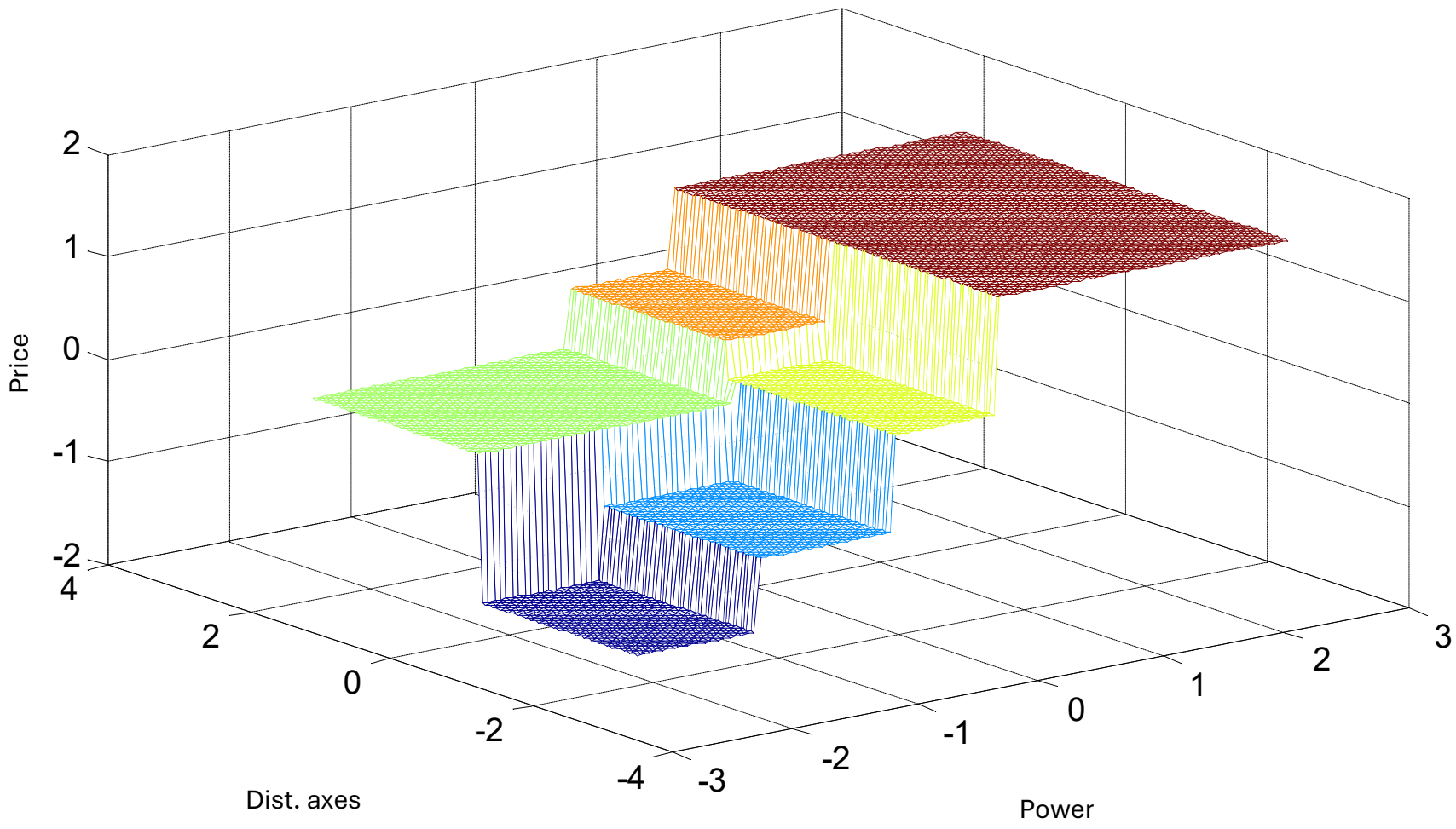
Regression Trees

- Regression trees:



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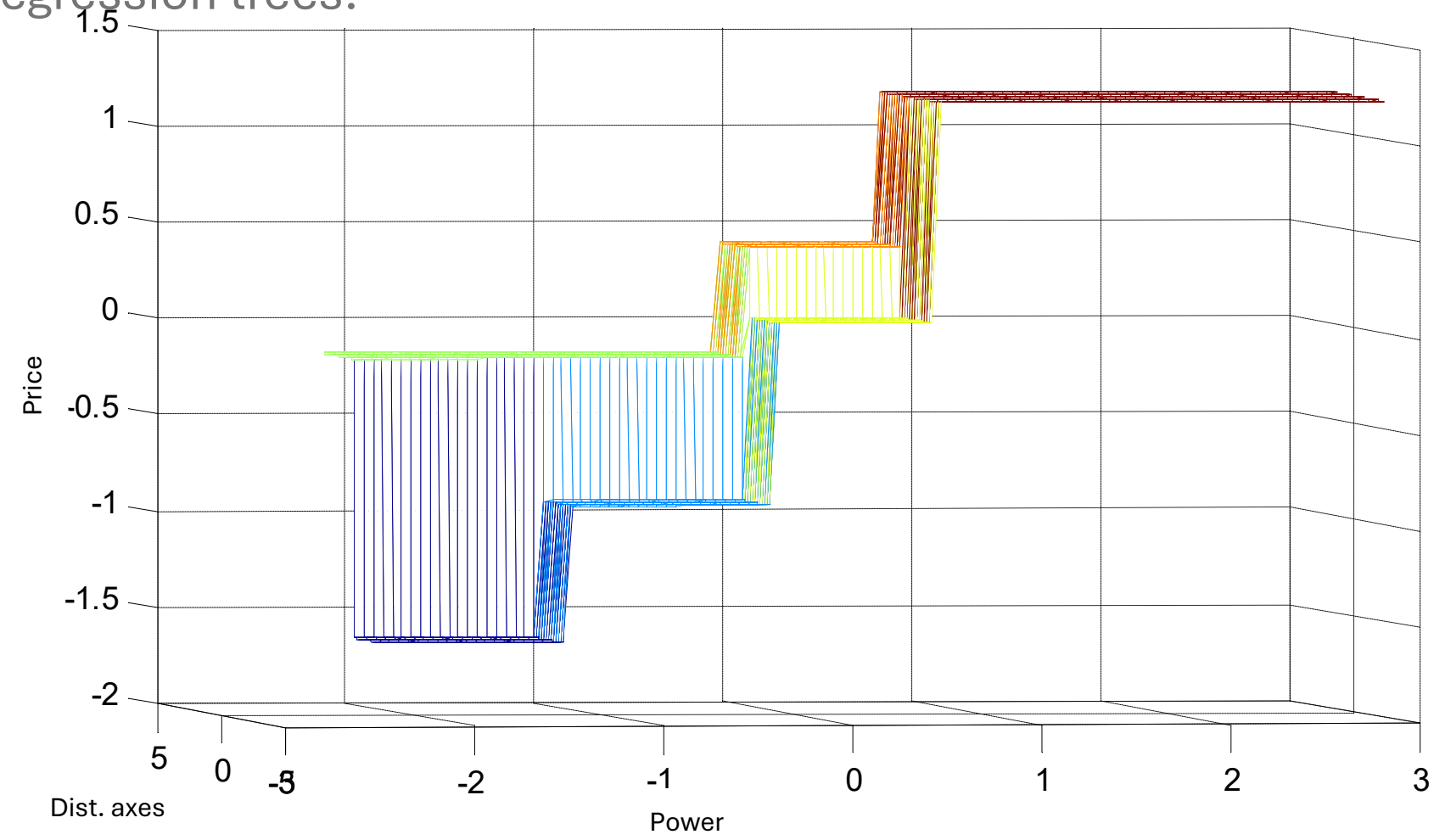
Regression Trees

- Regression trees:



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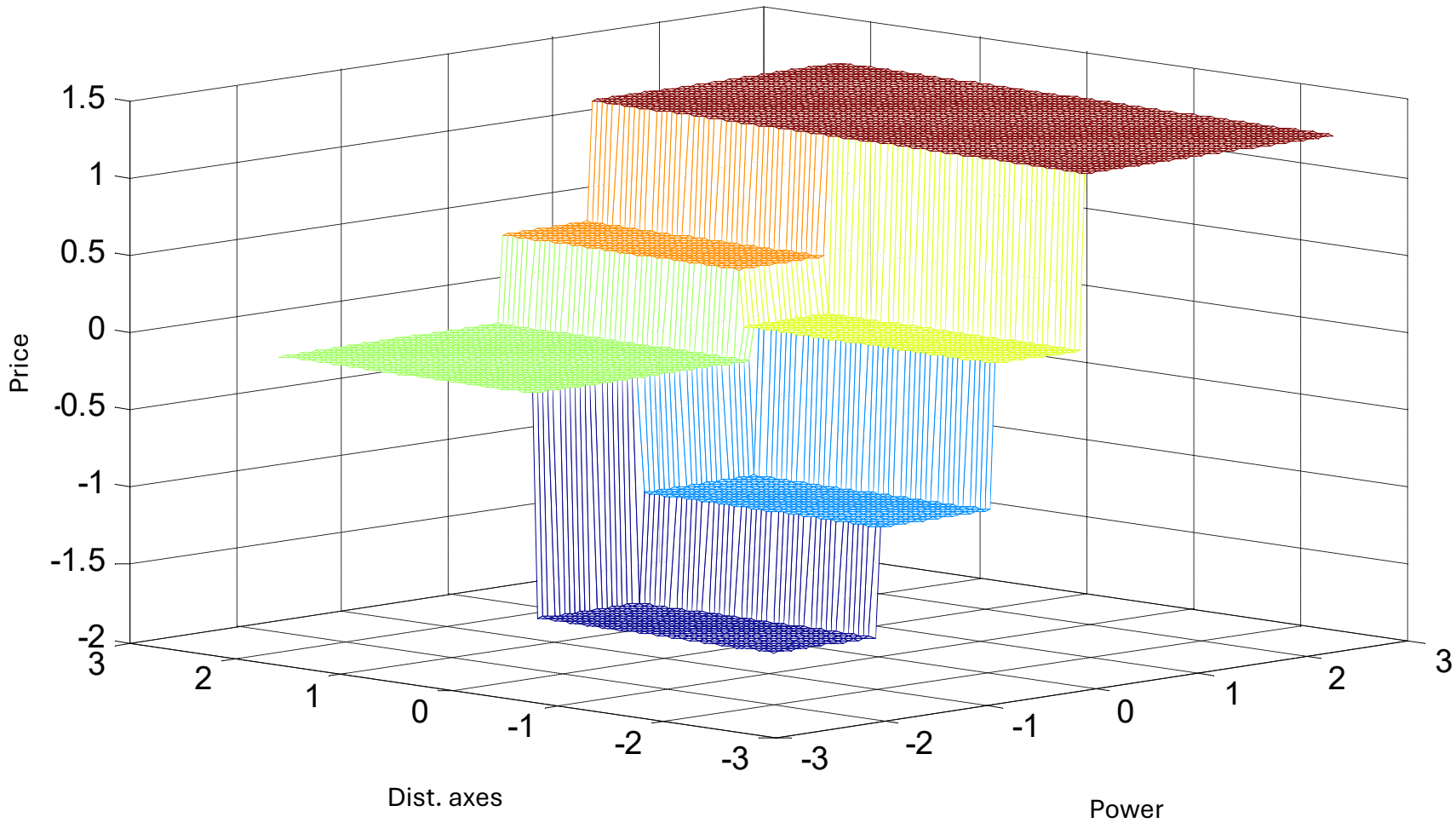
Regression Trees

- Regression trees:



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Regression Trees

- Regression trees:
  - Database with 209 different computer configurations

	Cycle time (ns) MYCT	Main memory (KB) Min. MMIN	Main memory (KB) Max. MMAX	Cache (KB) CACH	Channels Min. CHMIN	Channels Max. CHMAX	Performance PRP
1	125	256	6000	256	16	128	198
2	29	8000	32000	32	8	32	269
3	29	8000	32000	32	8	32	220
4	29	8000	32000	32	8	32	172
5	29	8000	16000	32	8	16	132
. . .							
207	125	2000	8000	0	2	14	52
208	480	512	8000	32	0	0	67
209	480	1000	4000	0	0	0	45

  - Output: processing power
  - All attributes, and output, are continuous

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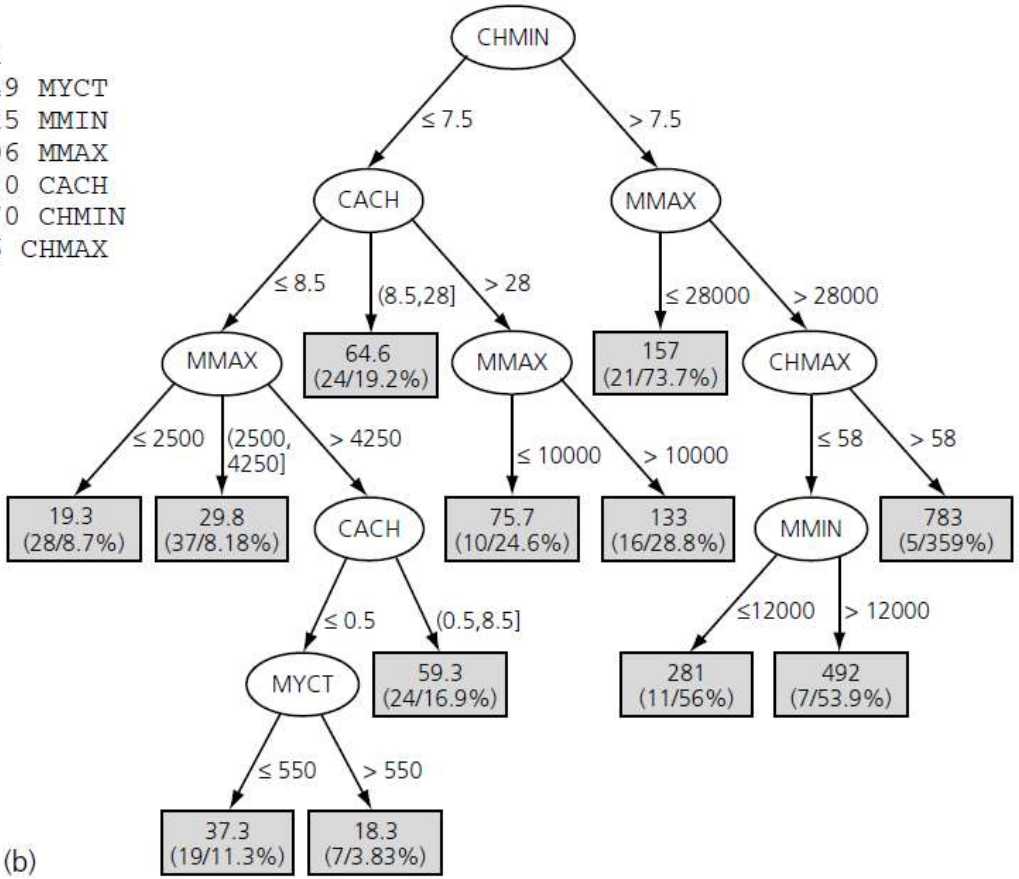
Regression Trees

- Regression trees:
  - Regression equation (linear) (a) and Regression tree (b)

```
PRP =
  -56.1
  +0.049 MYCT
  +0.015 MMIN
  +0.006 MMAX
  +0.630 CACH
  -0.270 CHMIN
  +1.46 CHMAX
```
(a)



(b)

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Regression Trees

- Regression trees:
  - Advantages:
    - Making predictions is fast
    - It is easy to understand which variables are important when making the prediction
    - If any data is missing,
      - It may not be possible to go all the way down the tree to the leaf
      - Even so, a prediction can be made by averaging all the leaves of the subtree arrived at
      - For example:

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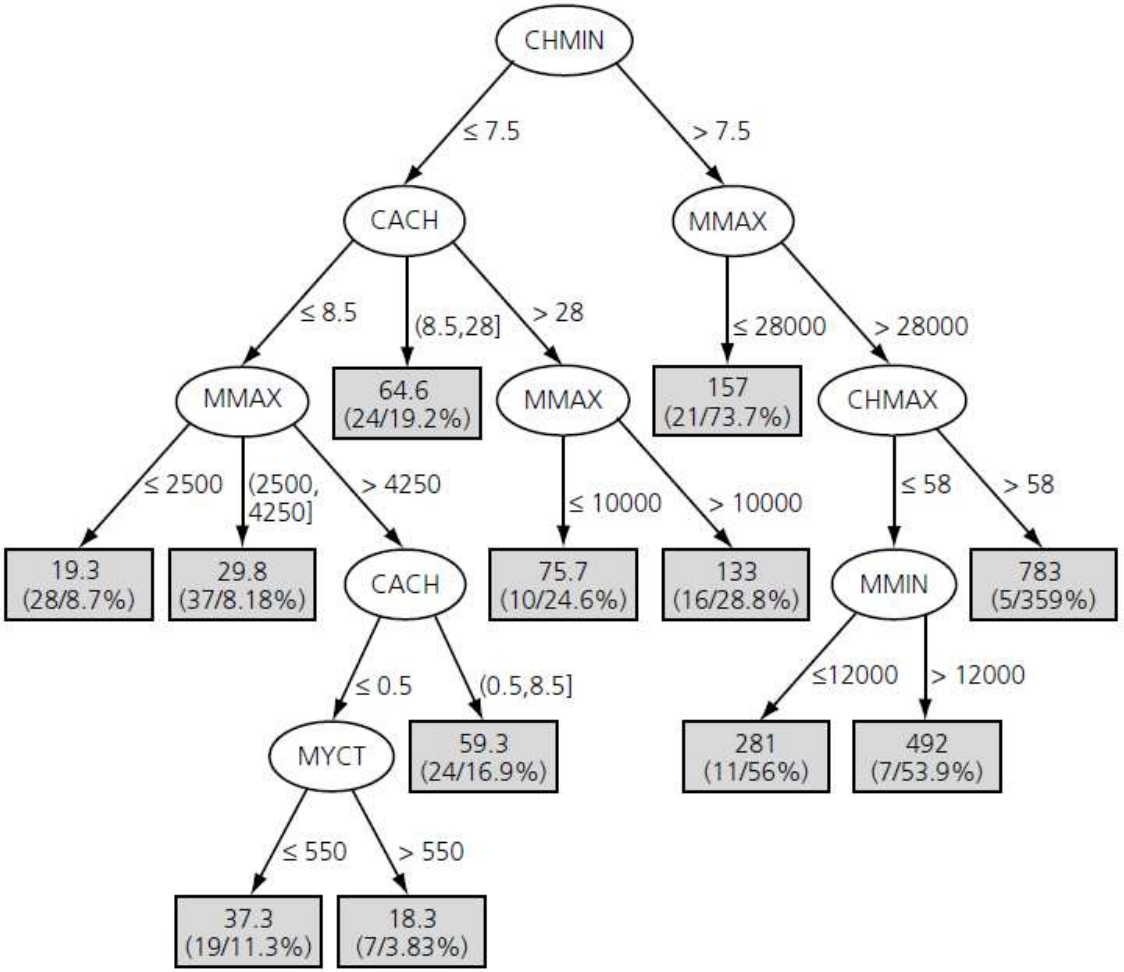
Regression Trees

- Regression trees:
  - Example:
    - CHMIN = 8
    - CHMAX = ?
    - MMIN = 8000
    - MMAX = 32000



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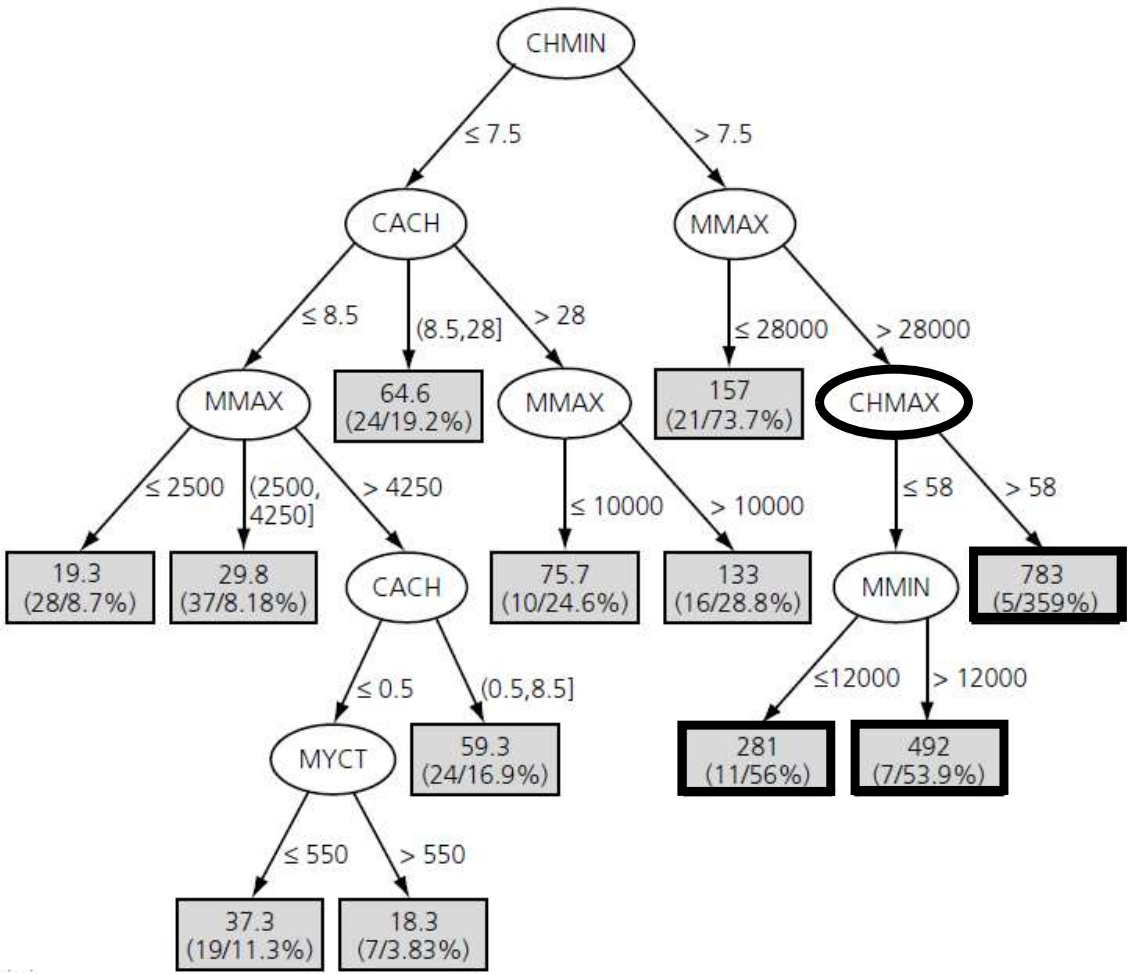
Regression Trees

- Regression trees:
  - Example:
    - CHMIN = 8
    - CHMAX = ?
    - MMIN = 8000
    - MMAX = 32000
  - Prediction:
    - (281+492+783) / 3



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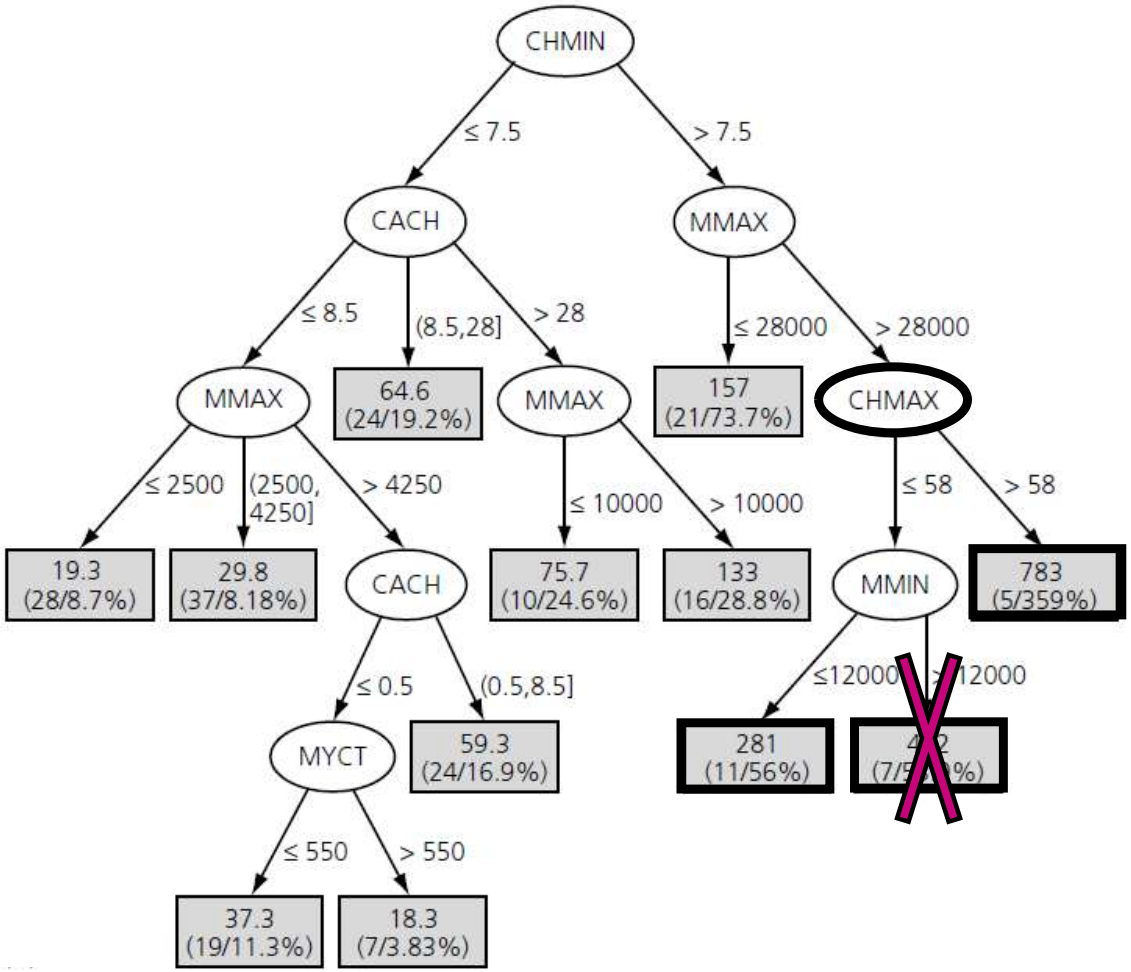
Regression Trees

- Regression trees:
  - Example:
    - CHMIN = 8
    - CHMAX = ?
    - MMIN = 8000
    - MMAX = 32000
  - Prediction:
    - ~~(28[illegible]/3~~



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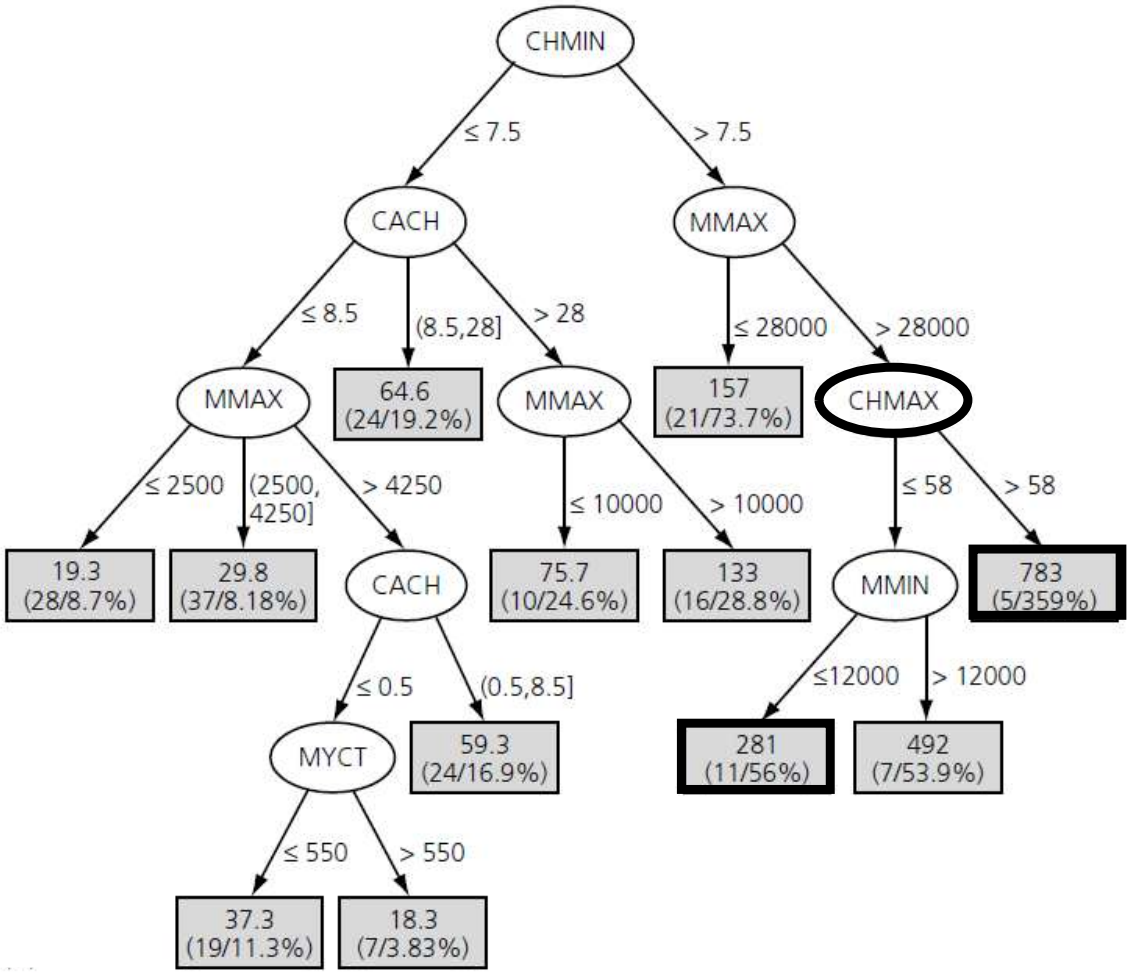
Regression Trees

- Regression trees
  - Example:
    - CHMIN = 8
    - CHMAX = ?
    - MMIN = 8000
    - MMAX = 3200
  - Prediction:
    - (281 + 783) / 2



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Regression Trees

- Regression trees:
  - Advantage:
    - Making predictions is fast
    - It is easy to understand which variables are important when making the prediction
    - If any data is missing,
      - It may not be possible to go all the way down the tree to the leaf
      - Even so, a prediction can be made by averaging all the leaves of the subtree arrived at
    - The model gives a step response
      - Can be applied when the desired regression surface is smooth
      - Even so, if it is not smooth, the piecewise constant surface can approximate it arbitrarily well
        - With a sufficient number of leaves
    - There are fast and efficient algorithms to develop these trees

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Regression Trees

- Regression model trees:
  - Leaves:
    - They contain a linear regression model to predict the response of the instances that arrive at that leaf
  - Algorithm M5 (Quinlan, 1992) and M5' (Wang and Witten, 1997)
    - M5 generates decision trees similar to ID3
    - Choose the attribute that minimizes the variation in class values that results from each branch.
    - M5' introduces improvements to:
      - Use discrete attributes
      - Handle missing values
      - Reduce the size of the tree

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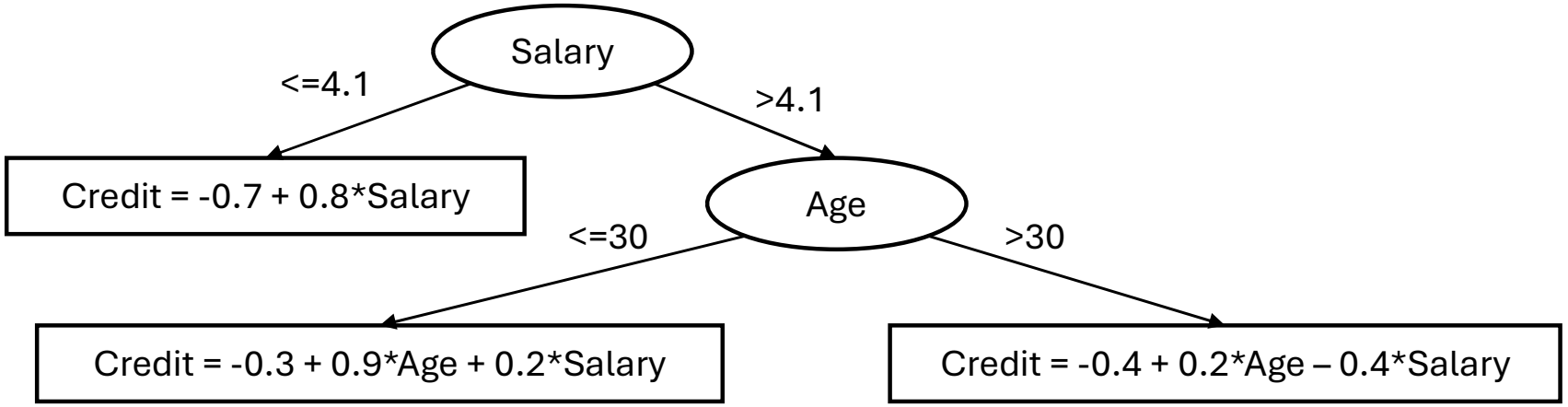
Regression Trees

- Example:

Client	Salary	Age	Children	...	Credit
1	4.5	3. 4	1	...	**4.53**
2	2.3	27	2	...	**1.16**
3	9.5	51	2	...	**6.21**
4	1.2	29	3	...	**0.13**
...	...	...	...	...	



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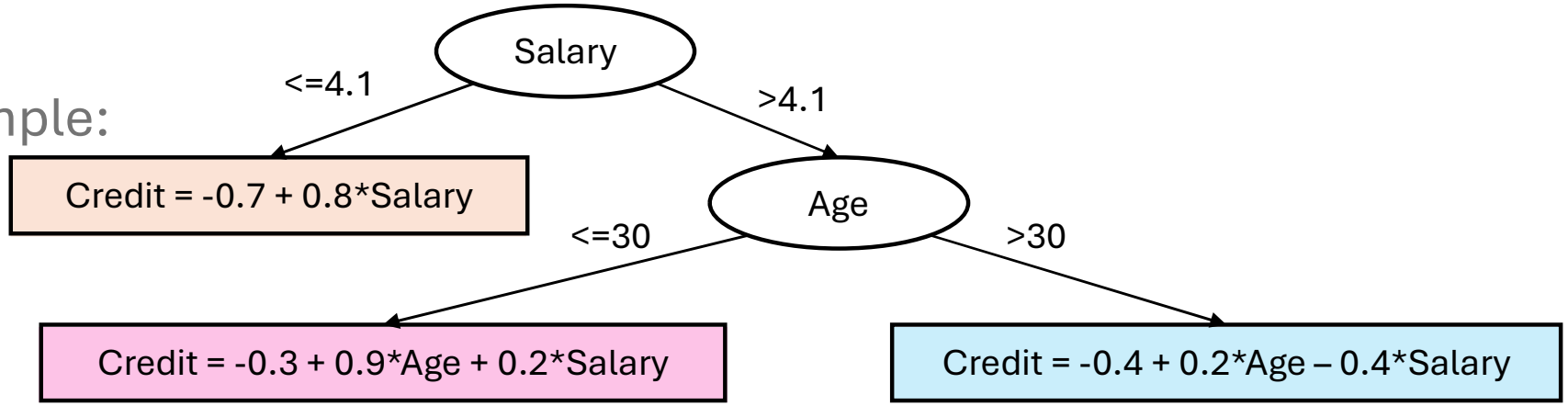
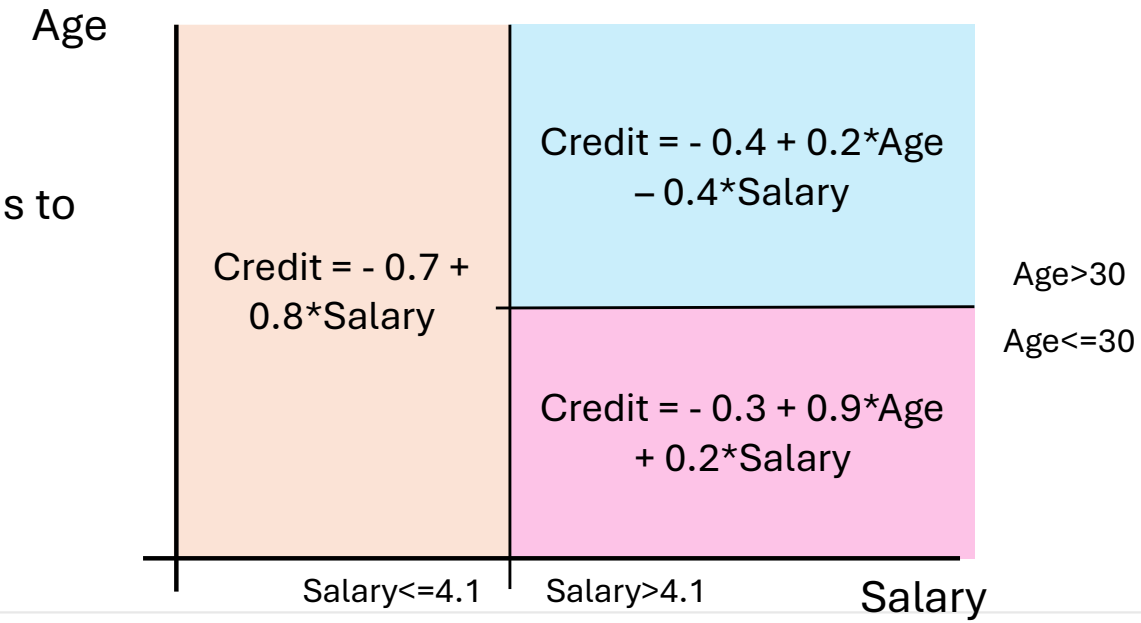
Regression Trees

- Example:



Each linear expression applies to a different region: **Local models**



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Regression Trees

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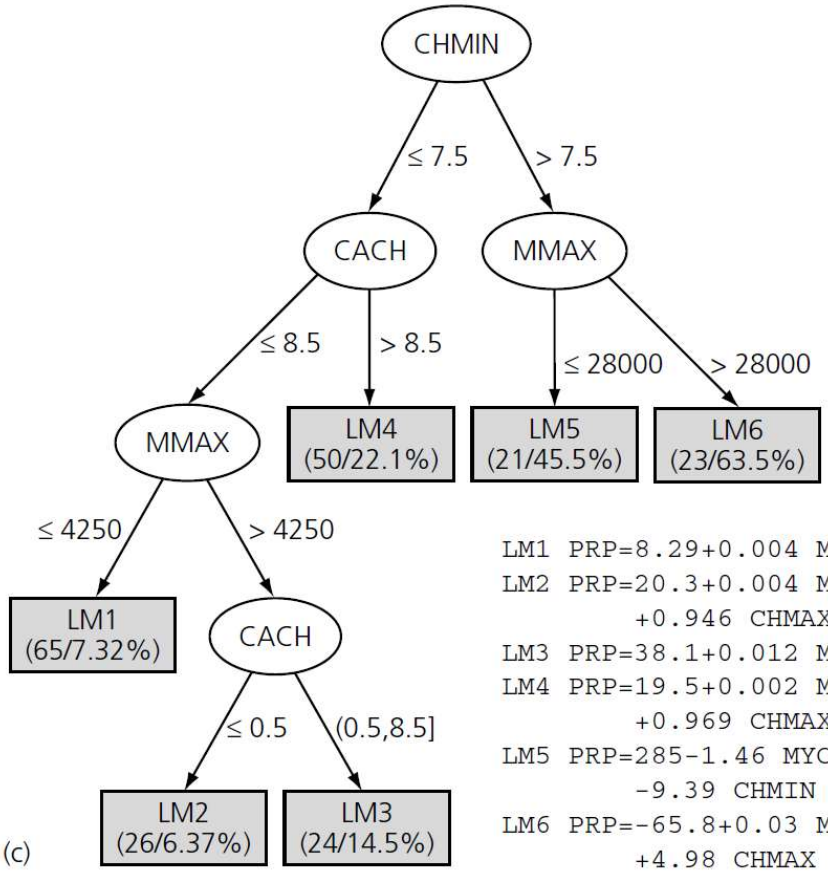
Regression Trees

- Example:
  - Regression Model Tree:




```
LM1 PRP=8.29+0.004 MMAX+2.77 CHMIN
LM2 PRP=20.3+0.004 MMIN-3.99 CHMIN
        +0.946 CHMAX
LM3 PRP=38.1+0.012 MMIN
LM4 PRP=19.5+0.002 MMAX+0.698 CACH
        +0.969 CHMAX
LM5 PRP=285-1.46 MYCT+1.02 CACH
        -9.39 CHMIN
LM6 PRP=-65.8+0.03 MMIN-2.94 CHMIN
        +4.98 CHMAX
```

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